Preregistration

Quantitative Judgments of Realistic Stimuli

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Study Information

Research Question

Which cognitive model of quantitative judgments best describes participants judgments in a classic multiple-cue judgment experiment with realistic stimuli from three different domains?

Description

Using a classical multiple-cue judgment experimental paradigm, in which participants first learn the criterion values of set of exemplars and then later judge a set of new stimuli, we are interested in which of five cognitive process models best describes participants judgments of realistic real-word stimuli and criteria. As real-world domains and judgment criteria we will use life expectancy of countries, days until female maturity of mammals, and amount of carbohydrates per 100g of food items. The models we will compare are an exemplar-based model (e.g. Izydorczyk & Bröder, 2021; Juslin et al., 2003; 2008; Juslin & Persson, 2002), a rule-based model (Juslin et al., 2003; 2008), the RulEx-J model (Bröder et al., 2017; Izydorczyk & Bröder, 2022), the mapping model (Helversen & Rieskamp, 2008, 2009), the *QuickEst* Heuristic (Hertwig, Hoffrage, & Martignon, 1999; Hausmann et al., 2007), as well as one random guessing model as baseline.

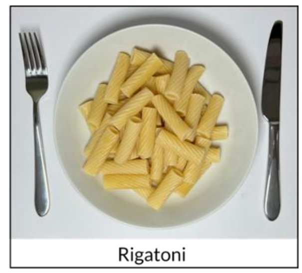
Exploratory Hypotheses

We will explore which models best describes the actual judgment data.

Study

Materials

There will be **80 items** from each of three different domains: countries, mammals, and food items. Examples stimuli are shown here:

For each domain, **12** from the 80 items will be selected as **exemplar or training items.** These exemplar items will be selected based on a simulation procedure, which in 100,000 iterations randomly selects 12 out of the 80 different items and then uses these 12 randomly selected items to make predictions for all items based on the rule, exemplar and mapping model, since these are the main competitor models and the RulEx-J model is a blending between rule and exemplar models. The exemplars will then be selected based on how different the predictions of the models are (i.e., how good the models can be distinguished based on these exemplars) as indicated by the Root-Mean-Squared Error (RMSE) between the pairwise model predictions, as well as the RMSE between the models and the actual values, since each model should also be able to predict the actual values to some degree, based on the given exemplars.

Procedure

The general procedure is based on classical multiple cue-judgment studies (e.g., Trippas & Pachur, 2019; Juslin et al., 2003; Bröder & Gräf, 2018). The study will consist of *two* phases, a **training phase** and a **testing phase**.

In one block of the training phase, participants will have to judge the criterion value of each of the 12 exemplars in a randomized order and then receive feedback about the true value directly afterwards. There will be a minimum of five training blocks up to 10 training blocks, depending on the performance of the participant. If after the fifth block, participants judge at least 10 of the 12 exemplars correctly (i.e., judgment is within a +- 10% interval of the true value), the training phase will end for them. In the testing phase, participants then judge all 80 items in a randomized order and there will be no feedback. In the testing phase there will also be three instructed attention check trials, where instead of an actual item from the domain, a basketball will be shown, and participants are instructed to answer with "999". Afterwards, there will be some demographic questions and questions about compliance and potential problems during the study.

The domain (countries, mammals, food) will be a between-subjects factor, this is, each participants will judge only one set of items.

Variables

Measured variables

- Consent (yes/no)
- Judgments
- Response time (time in ms until first input is made)

Participants

Data collection procedures

Participants will be recruited through Prolific and will be paid £5 for the 30 min study. Recruitment criteria in Prolific are that participants have to be a least 18 years old to participate, speak English fluently and have at least a 95% approval rating on Prolific.

Sample size We will collect $N = 150$ participants in total, i.e., $n = 50$ for each of the three domains.

Sample size rationale Based on previous studies, $N = 50$ participants are enough for model comparison.

Analysis Plan

Based on the findings of Izydorczyk & Bröder (2021), we will use only the judgments for the 68 new stimuli (i.e., excluding the exemplars) of each participant in the data analysis. Model comparison will be done using BayesFlow framework for amortized Bayesian inference (Radev et al., 2023), where we will compare Bayesian Hierarchical implementations of each of the five models using Bayes Factors.

Data exclusion

Participants will be excluded …

- … if the mean response times in the testing phase is 3 SDs below the sample
- … if they fail one of the three attention checks
- … if they make only two or less different estimates for all items in the testing phase.
- … if they indicated non-compliance. That means they either indicate having cheated and or just having clicked through the experiment (Aust et al., 2013)
- … if they indicated that they used some aid (looking up answers, taking notes, asking other persons)
- … if they indicated and described technical problems or that the experiment did not work as intended.
- … if they provided judgments in under 1000 ms for more than 10% of the items in the estimation phase